



- 6** The curve C has polar equation $r = \cos \frac{1}{2}\theta$, for $0 \leq \theta \leq \pi$.
- (a)** Sketch C .

[2]

In parts **(b)** and **(c)** you may use the identities $\sin \theta \equiv 2 \sin \frac{1}{2} \theta \cos \frac{1}{2} \theta$ and $\cos \theta \equiv 2 \cos^2 \frac{1}{2} \theta - 1$.

- (b) Find the exact value of the area of the region enclosed by C and the initial line. [4]

[illegible]



- (c) Find the maximum distance of a point on C from the initial line. Give your answer in the form $p\sqrt{q}$, where p and q are rational numbers to be determined. [6]

[illegible]